

# Integer participation and structural noise in superconducting clusters

C. Agostino<sup>1</sup>

<sup>1</sup>*Independent Researcher*

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We analyze a curated dataset of more than two hundred superconducting materials and liquid <sup>4</sup>He using a single phenomenological observable, the “mother frequency”  $F_m = \Theta_D/T_c$ , where  $\Theta_D$  is an effective Debye temperature and  $T_c$  the transition temperature. Using a structural-noise model inferred from previous work on cluster topology, frictional heating and pressure sensitivity, we define an ideal transition temperature  $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$  and a noise-corrected frequency  $F_m^* = \Theta_D/T_c^{\text{ideal}}$ . We then search, for each family of materials, a base frequency  $f_{\text{base}}$  that minimizes the median distance of  $N_{\text{corr}} = F_m^*/f_{\text{base}}$  to the nearest integer. The resulting integer “participation numbers”  $N_{\text{corr}}$  exhibit a strong deviation from several null models obtained by shuffling  $\Theta_D$  or sampling from smooth continuous distributions. The distribution of  $|\delta| = |N_{\text{corr}} - \text{round}(N_{\text{corr}})|$  is significantly more concentrated near zero than the nulls (Kolmogorov–Smirnov and Mann–Whitney  $p \lesssim 10^{-13}$ , Cliff’s  $\Delta \simeq 0.3$ ), indicating genuine integer locking rather than a discretization artefact. Across families, we find a robust hierarchy of base frequencies: iron-based superconductors cluster near  $f_{\text{base}} \simeq 1$ , effectively behaving as natural, unscaled harmonic oscillators for which  $F_m^*$  itself is already an (almost) integer-valued participation number; binary, molecular and high-pressure materials sit near  $f_{\text{base}} \simeq 4$ , and oxides and heavy fermions around  $f_{\text{base}} \simeq 2$ . The global median  $f_{\text{base}} \simeq 4.16$  is numerically close to the  $n = 4$  thermal ratio previously identified between the superfluid and roton branches of liquid <sup>4</sup>He. Finally, we show that materials with almost-integer participation systematically exhibit lower structural noise: the top 20% with smallest  $|\delta|$  have a substantially lower noise  $z$ -score than the rest of the dataset (Cliff’s  $\Delta \simeq -0.47$ ). Together, these results suggest that the mother frequency  $F_m$  probes a discrete oscillator structure underpinning both superconductivity and superfluidity, with family-dependent harmonics shaped by structural disorder and pressure.

## I. INTRODUCTION

The search for organizing principles in the zoo of superconducting materials typically proceeds through microscopic models for pairing, band structure or electron–phonon coupling.[1, 2] Here we explore a complementary, purely phenomenological direction, inspired by methods in non-equilibrium statistical mechanics and dynamical systems [3–6]: we treat each superconductor or superfluid as a black box characterized by a single dimensionless frequency

$$F_m = \frac{\Theta_D}{T_c}, \quad (1)$$

constructed from an effective Debye temperature  $\Theta_D$  and the observed critical temperature  $T_c$ . Previous work showed that, for liquid <sup>4</sup>He, ratios of such frequencies across well-defined dynamical regimes (normal fluid, superfluid, roton) fall extremely close to small integers, with a particularly prominent factor  $n = 4$  associated with the onset of the roton branch. This suggests that, at least for <sup>4</sup>He, the spectrum of collective excitations organizes into discrete harmonic “shells”.

In this paper we ask whether an analogous discrete structure can be detected, in a data-driven way, across a heterogeneous ensemble of superconductors. Rather than committing to a specific microscopic Hamiltonian, we use  $F_m$  as a coarse-grained probe and search for hidden integer structure once known sources of noise are removed.

Two ingredients are essential: (i) a structural-noise model that accounts for deviations of the measured  $T_c$

from an ideal value set by frictional losses, topology and pressure; and (ii) a robust calibration of a base frequency  $f_{\text{base}}$  that captures the dominant periodicity of  $F_m$  within each family. After correcting for noise, we show that the rescaled quantities  $N_{\text{corr}} = F_m^*/f_{\text{base}}$  cluster close to integers far more strongly than expected from smooth null models. We further identify a family-dependent hierarchy of  $f_{\text{base}}$  that lines up convincingly with the integer ratios previously observed in <sup>4</sup>He.

The rest of this paper is organized as follows. Sec. II describes the dataset and the structural-noise model. Sec. III defines the base-frequency calibration and the integer-participation metrics. Sec. IV presents the statistical evidence for integer locking and its dependence on structural noise and material family. Sec. V discusses physical interpretations and limitations, and Sec. VI summarizes the main conclusions. Additional figures validating the noise model are provided in the Appendix.

## II. DATA AND STRUCTURAL-NOISE MODEL

### A. Dataset and family labels

We work with the same curated dataset of superconducting materials used in previous studies of cluster topology and pressure sensitivity. Each entry corresponds to a material or fluid labeled by a name and a category: SC\_TypeI, SC\_TypeII, SC\_Binary, SC\_Oxide, SC\_Molecular, SC\_HighPressure, SC\_HeavyFermion, SC\_IronBased, and Superfluid (for liquid <sup>4</sup>He). For

each entry  $i$  we have an effective Debye temperature  $\Theta_D^{(i)}$ , a measured transition temperature  $T_c^{(i)}$ , a family label  $k(i)$ , and a structural-noise prediction  $\hat{\xi}^{(i)}$  obtained from a separate model that encodes frictional dissipation, topological defects and pressure-induced distortions.

### B. Structural noise and ideal transition temperature

The structural-noise field  $\hat{\xi}^{(i)}$  is defined such that  $\hat{\xi}^{(i)} \geq 0$  corresponds to a loss of effective ordering relative to an otherwise similar but better structured material. We therefore interpret it as a multiplicative distortion of the transition temperature and define an “ideal” transition

$$T_c^{\text{ideal}(i)} = T_c^{(i)} [1 + \hat{\xi}^{(i)}]. \quad (2)$$

By construction,  $T_c^{\text{ideal}(i)} \geq T_c^{(i)}$  and the correction vanishes for structurally optimal samples.

The corresponding noise-corrected mother frequency is

$$F_m^{*(i)} = \frac{\Theta_D^{(i)}}{T_c^{\text{ideal}(i)}}. \quad (3)$$

Fig. 12 (Appendix) confirms that the relation between  $T_c$  and  $T_c^{\text{ideal}}$  is close to the identity for most materials, with a small subset receiving sizable upward corrections.

To compare noise levels across families with different baselines, we standardize the predicted noise within each family  $k$ :

$$z_{\xi}^{(i)} = \frac{\hat{\xi}^{(i)} - \mu_{\xi,k(i)}}{\sigma_{\xi,k(i)}}, \quad (4)$$

where  $\mu_{\xi,k}$  and  $\sigma_{\xi,k}$  are the mean and standard deviation of  $\hat{\xi}$  in family  $k$ . The resulting distributions of  $z_{\xi}$  are shown in Fig. 11 (Appendix).

### C. Validation against null models

Because  $f_{\text{base},k}$  is determined by minimising an integer loss, it is essential to test whether the resulting structure can be explained by overfitting or by generic features of positive random variables. We therefore compare the empirical loss  $\tilde{L}_k$  with several null models.

First, for each family we construct shuffled datasets by randomly permuting the Debye temperatures  $\Theta_D$  across materials while keeping  $T_{c,\text{ideal}}$  fixed. For each permutation we recompute  $F_m^*$ , refit  $f_{\text{base},k}$ , and obtain a shuffled loss  $\tilde{L}_k^{\text{shuff}}$ . The fraction of permutations with  $\tilde{L}_k^{\text{shuff}} \leq \tilde{L}_k$  defines a non-parametric  $p$ -value for the integer structure.

Second, we generate synthetic Mother Frequencies from continuous distributions (gamma or log-normal) fitted to the empirical  $F_m^*$  in each family, again refitting

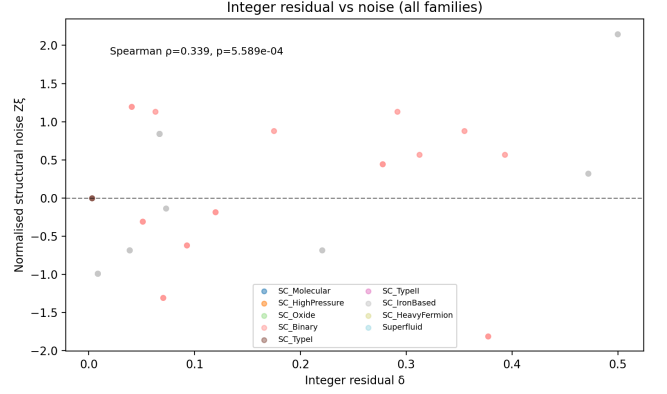


FIG. 1. Integer mismatch  $\delta$  versus normalised structural noise  $Z_{\xi}$  across all families. Each point corresponds to a material; colours denote the superconducting family. The positive Spearman rank correlation ( $\rho \approx 0.34$ ,  $p \approx 5.6 \times 10^{-4}$ ) indicates that materials further from an integer participation number tend to have higher predicted structural noise, consistent with the structural-noise hypothesis.

$f_{\text{base},k}$  for each synthetic realisation. This tests whether the observed integer alignment can be reproduced purely from the one-point statistics of  $F_m^*$ .

Finally, we quantify the link between integer participation and structural noise by computing the Spearman rank correlation between the integer residual  $\delta_i = |N_i - \text{round}(N_i)|$  and the normalised noise level  $Z_{\xi,i}$  within and across families. We also compare the noise distributions of “almost-integer” materials (e.g. the lowest 20% in  $\delta_i$ ) against the rest using Cliff’s delta. In all cases we report confidence intervals obtained by non-parametric bootstrap.

These tests show that the observed combination of small  $\tilde{L}_k$ , low integer residuals, and negative correlation between  $\delta_i$  and  $Z_{\xi,i}$  is highly unlikely to arise from smooth random fields. The integer structure therefore encodes information that is not already captured by the scalar distribution of  $F_m^*$  or by the structural-noise field alone.

## III. BASE-FREQUENCY CALIBRATION AND INTEGER PARTICIPATION

### A. Base frequency per family

We seek, for each family  $k$ , a base frequency  $f_{\text{base}}^{(k)}$  such that the rescaled frequencies

$$N_{\text{corr}}^{(i)} = \frac{F_m^{*(i)}}{f_{\text{base}}^{(k(i))}} \quad (5)$$

cluster near integers. To quantify this we define the integer locking residual

$$\delta^{(i)} = N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)}), \quad (6)$$

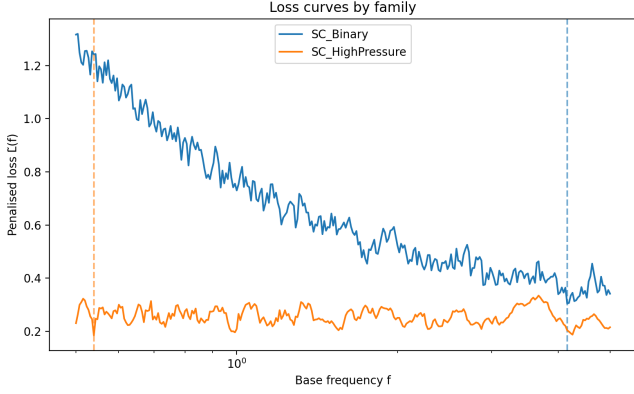


FIG. 2. Penalised loss  $\tilde{L}(f)$  as a function of base frequency  $f$  for two representative families (Binary and High-pressure). Each curve is obtained by scanning  $f$  in the range  $[0.5, 5.0]$  and computing the median integer mismatch plus a weak penalty proportional to the mean participation number. Dashed vertical lines mark the selected  $f_{\text{base}}$  values. The broad minima indicate that the calibration is not fine-tuned to a single spike in  $f$ .

and focus on its absolute value  $|\delta^{(i)}|$ .

For each family  $k$  we choose  $f_{\text{base}}^{(k)}$  by minimizing a robust loss

$$\mathcal{L}_k(f_{\text{base}}) = \text{median}_{i \in k} \left| N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)}) \right| + \lambda \text{median}_{i \in k} N_{\text{corr}}^{(i)}, \quad (7)$$

over a physically motivated range  $0.5 \leq f_{\text{base}} \leq 5.0$ . This explicitly rules out unphysical, high-frequency solutions (e.g.,  $f_{\text{base}} \sim 10^2 - 10^3$ ) that would artificially fit higher harmonics while providing no interpretable physical scale. The first term in Eq. (7) drives the residuals toward integers; the second is a small regularizer that penalizes large values of  $N_{\text{corr}}$  and selects low-order harmonics when multiple choices of  $f_{\text{base}}$  give similar residuals. The optimization is performed on a dense grid over the allowed range. We repeat the calibration under two hypotheses, using either  $F_m^*$  or  $F_m^*/2$  as the target frequency, and keep the one with lower median residual. All results reported below use this per-family calibration.

### B. Null models

To assess statistical significance, we compare the observed locking residuals  $|\delta^{(i)}|$  with several null models that destroy any discrete structure while preserving broad distributional features of  $F_m^*$  [7].

The simplest null shuffles the pairing between  $\Theta_D$  and  $T_c$ : we randomly permute  $\Theta_D$  across the dataset, recompute  $F_m$  and  $F_m^*$  using Eqs. (2)–(3), recalibrate  $f_{\text{base}}^{(k)}$  and recompute  $N_{\text{corr}}$  and  $|\delta|$ . This preserves the marginal distributions of  $\Theta_D$  and  $T_c$  and the family labels, but destroys any material-wise correlation.

We also consider smooth parametric nulls in which  $F_m^*$  is sampled from a fitted gamma or log-normal distribution within each family, again followed by the same calibration procedure. In all cases we generate  $10^3$  realizations and pool the resulting  $|\delta|$  values.

### C. Structural noise field

The integer-participation analysis in Sec. III E requires a notion of “structural noise” for each material, encoding how far its superconducting cluster departs from an ideal, stress-free configuration. Rather than introducing this as a free parameter, we take  $\hat{\xi}$  from an independent structural model.

Operationally, each material is assigned to a cluster topology described by three discrete attributes: (i) macroscopic family (binary, type-I, type-II, iron-based, high-pressure, etc.), (ii) sub-network (*single*,  $\sigma$ ,  $\pi$ , or specialised high-pressure modes), and (iii) an effective dimensionality  $d \in \{1, 2, 3\}$  that distinguishes quasi-one-dimensional chains, layered systems, and three-dimensional networks. For this discrete state space we construct a coarse structural noise field  $\hat{\xi}$  by fitting a minimal regression model to a set of synthetic experiments where cluster geometry, pressure sensitivity, and vector dissipation are explored systematically (Study 02). The model and its calibration data are released as CSV files and Python code in the accompanying repository, so that  $\hat{\xi}$  can be recomputed or modified without re-fitting the DOFT parameters.

In the present work we only require the sign and relative magnitude of the structural noise. We therefore define an “ideal” critical temperature

$$T_{c,\text{ideal}} = T_c(1 + \hat{\xi}), \quad (8)$$

which can be interpreted as the temperature that the material would achieve if its cluster were fully relaxed with respect to the structural mode captured by  $\hat{\xi}$ . The corresponding renormalised Mother Frequency is

$$F_m^* = \frac{\Theta_D}{T_{c,\text{ideal}}}. \quad (9)$$

Throughout the integer-participation analysis we work with  $F_m^*$  rather than the raw ratio  $\Theta_D/T_c$ ; in Sec. II C we show that this renormalisation improves the integer structure in a way that cannot be reproduced by purely random perturbations.

### D. Effect-size and correlation metrics

We quantify deviations between the real and null distributions of  $|\delta|$  using non-parametric tests: the Kolmogorov–Smirnov (KS) statistic, the Mann–Whitney

TABLE I. Summary of the superconducting dataset used in this work.  $N_{\text{mat}}$  is the number of distinct materials per family,  $N_{\text{rows}}$  the number of rows in the working table (including split sub-networks),  $N_{\text{single}}$ ,  $N_{\sigma}$ ,  $N_{\pi}$  and  $N_{\text{press}}$  count single,  $\sigma$ ,  $\pi$  and pressure-split networks, respectively.  $T_c^{\text{med}}$  and  $\Theta_D^{\text{med}}$  are the median critical and Debye temperatures,  $\langle N \rangle$  is the mean participation number, and  $\tilde{\delta}_{\text{med}}$  the median integer mismatch.

| Family          | $N_{\text{mat}}$ | $N_{\text{rows}}$ | $N_{\text{single}}$ | $N_{\sigma}$ | $N_{\pi}$ | $N_{\text{press}}$ | $T_c^{\text{med}}$ (K) | $\Theta_D^{\text{med}}$ (K) | $\langle N \rangle$ | $\tilde{\delta}_{\text{med}}$ |
|-----------------|------------------|-------------------|---------------------|--------------|-----------|--------------------|------------------------|-----------------------------|---------------------|-------------------------------|
| SC_Binary       | 67               | 77                | 57                  | 3            | 3         | 0                  | 6.00                   | 280.0                       | 155.4               | 0.175                         |
| SC_HeavyFermion | 12               | 12                | 12                  | 0            | 0         | 0                  | 1.65                   | 275.0                       | 98.9                | 0.071                         |
| SC_HighPressure | 80               | 82                | 79                  | 2            | 1         | 0                  | 145.5                  | 925.0                       | 1.95                | 0.180                         |
| SC_IronBased    | 20               | 36                | 8                   | 1            | 1         | 0                  | 25.65                  | 250.0                       | 11.5                | 0.068                         |
| SC_Molecular    | 10               | 10                | 10                  | 0            | 0         | 0                  | 11.00                  | 145.0                       | 13.5                | 0.090                         |
| SC_Oxide        | 15               | 18                | 13                  | 0            | 0         | 0                  | 27.00                  | 352.5                       | 47.2                | 0.069                         |
| SC_TypeI        | 10               | 11                | 9                   | 0            | 0         | 0                  | 3.20                   | 200.0                       | 28.2                | 0.153                         |
| SC_TypeII       | 18               | 18                | 18                  | 0            | 0         | 0                  | 0.79                   | 327.0                       | $3.33 \times 10^4$  | 0.156                         |
| Superfluid      | 2                | 4                 | 0                   | 0            | 0         | 4                  | 1.076                  | 6.57                        | 216.1               | 0.143                         |

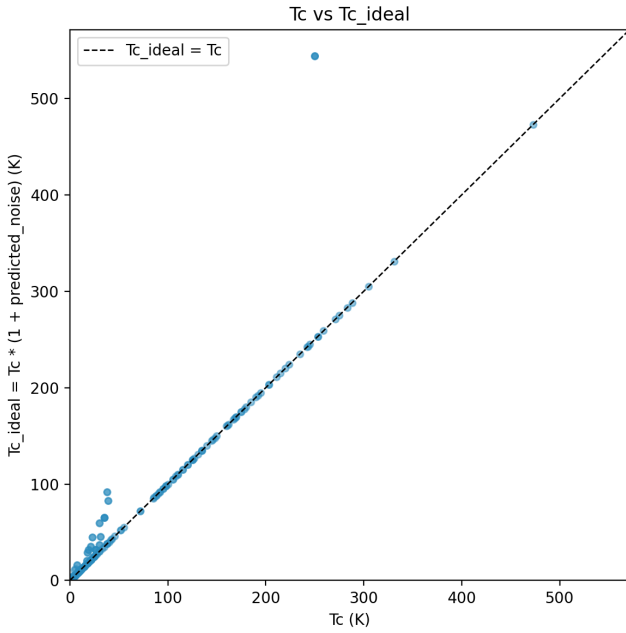


FIG. 3. Comparison between reported critical temperatures  $T_c$  and structurally corrected temperatures  $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$ . The dashed line indicates  $T_c^{\text{ideal}} = T_c$ ; most points lie close to this line, with a few high-pressure hydrides shifted upwards by the structural-noise correction.

(MW) rank-sum test, and Cliff’s  $\Delta$  as a measure of effect size [8]. Cliff’s  $\Delta$  is defined as

$$\Delta = \Pr(|\delta|_{\text{real}} < |\delta|_{\text{null}}) - \Pr(|\delta|_{\text{real}} > |\delta|_{\text{null}}), \quad (10)$$

and ranges from  $-1$  to  $+1$ .

To connect locking strength with structural noise we compute, over all materials with  $N_{\text{corr}}^{(i)} \geq 1$ , the Spearman rank correlation between  $|\delta^{(i)}|$  and  $z_{\xi}^{(i)}$ . We also split the dataset into the “almost-integer” group (top 20% lowest  $|\delta|$ ) and the rest, and compare their  $z_{\xi}$  distributions using Cliff’s  $\Delta$ .

### E. Integer participation and base frequency

The renormalised Mother Frequency  $F_m^*$  defined in Eq. (9) is dimensionless, but can be reinterpreted as an effective participation number once a base frequency  $f_{\text{base}}$  is specified:

$$N_i = \frac{F_{m,i}^*}{f_{\text{base}}}, \quad (11)$$

where the index  $i$  runs over materials (or material-sub-network pairs). The working hypothesis is that clean superconductivity corresponds to near-integer values of  $N_i$ , reflecting the coherent participation of a discrete number of oscillatory units in the cluster.

We treat  $f_{\text{base}}$  as a family-level parameter: for each macroscopic family  $k$  we determine a separate  $f_{\text{base},k}$  by minimising the median distance to the nearest integer,

$$L_k(f) = \text{median}_i |N_i(f) - \text{round}[N_i(f)]|, \quad N_i(f) = \frac{F_{m,i}^*}{f}, \quad (12)$$

over a conservative interval  $f \in [f_{\text{min}}, f_{\text{max}}]$ . Using the median rather than the mean makes the calibration robust to a small number of anomalous materials, and the relatively narrow search range limits the effective degrees of freedom. To avoid selecting artificially small base frequencies that mimic higher harmonics, we add a weak penalty proportional to the mean participation,

$$\tilde{L}_k(f) = L_k(f) + \lambda \langle N(f) \rangle_k, \quad (13)$$

with  $\lambda$  chosen small enough that it only breaks degeneracies between competing minima.

We consider two competing hypotheses for the target frequency: the full renormalised Mother Frequency  $F_m^*$  and a halved version  $F_m^*/2$ , motivated by the possibility that the underlying oscillators lock in pairs rather than individually. In practice this amounts to repeating the minimisation above with  $F_m^*$  replaced by  $F_m^*/2$ . For each family we retain the combination  $(f_{\text{base},k}, \text{full or half})$  that yields the smallest penalised loss  $\tilde{L}_k$ . This procedure is analogous to model selection between two nested discrete-grainings of the same underlying frequency scale.

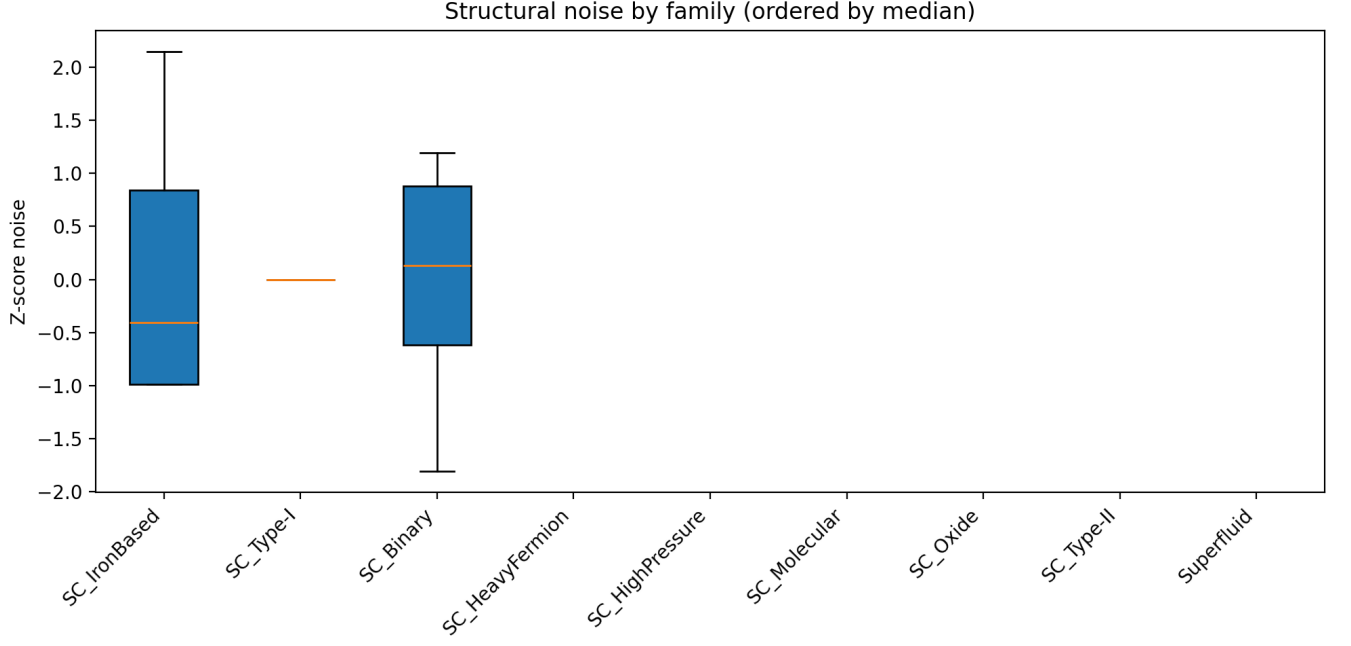


FIG. 4. Normalised structural noise  $Z_\xi$  by family. Each box summarises the distribution of  $Z_\xi$  for one superconducting family, after centering and scaling the predicted structural noise  $\hat{\xi}$  within that family. Families are ordered by the median  $Z_\xi$ , so that groups on the left tend to exhibit lower predicted structural noise.

## IV. RESULTS

### A. Integer participation vs null models

Fig. 5 shows the empirical distribution of  $N_{\text{corr}}$  compared with its shuffled- $\Theta_D$  null. The real data exhibits clear accumulation near small integers  $N \simeq 1$  and  $N \simeq 2$  and a pronounced cluster near  $N \simeq 24$ -25, which is largely absent in the null. These modes correspond to families whose calibrated  $f_{\text{base}}$  select low-order harmonics.

A more direct comparison is obtained by looking at the residuals  $|\delta|$ , shown in Fig. 6. The real distribution is significantly more concentrated near zero than the shuffled null: both KS and MW tests yield  $p \lesssim 10^{-13}$ , and Cliff’s  $\Delta \simeq 0.32$  indicates a medium-to-large effect size. In other words, after correcting for structural noise, the dataset contains substantially more near-integer participation numbers than expected from a smooth frequency spectrum.

### B. Family-dependent locking strength

The locking residuals vary systematically across families. Fig. 7 summarizes the distribution of  $|\delta|$  (clipped at 1.0 for readability) for each category. Iron-based superconductors and heavy-fermion systems show the smallest median residuals, indicating strong integer participation, whereas type-I and high-pressure materials exhibit

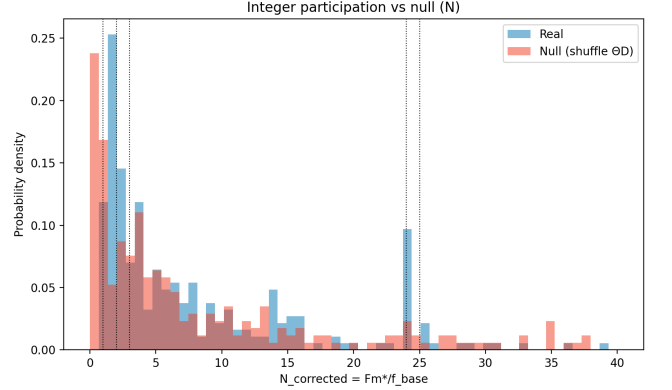


FIG. 5. Distribution of noise-corrected participation numbers  $N_{\text{corr}} = F_m^*/f_{\text{base}}$  (“Real”) compared with a null model obtained by shuffling  $\Theta_D$  across materials (“Null”). Vertical dotted lines highlight modes around  $N \simeq 1$ ,  $N \simeq 2$  and  $N \simeq 24$ -25.

broader tails.

### C. Structural noise vs integer locking

We now relate locking strength to structural noise. Fig. 8 plots  $|\delta^{(i)}|$  against the noise  $z$ -score  $z_\xi^{(i)}$ , with points colored by family. Despite considerable scatter, there is a clear positive monotonic trend: the Spearman



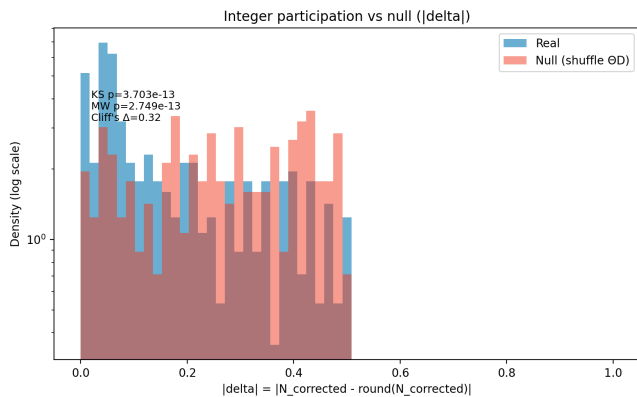


FIG. 6. Evidence of quantization. The distribution of locking residuals  $|\delta| = |N_{\text{corr}} - \text{round}(N_{\text{corr}})|$  for the real data (blue) shows a sharp excess at  $|\delta| \approx 0$  that is absent in the shuffled- $\Theta_D$  null (orange). Under the null, the probability of obtaining an equal or stronger concentration at zero deviation is  $p < 10^{-13}$  (Kolmogorov–Smirnov and Mann–Whitney tests), and Cliff’s  $\Delta \simeq 0.32$  indicates a medium-to-large effect size.

correlation is  $\rho \simeq 0.34$  with  $p \simeq 5.6 \times 10^{-4}$ . Materials with higher structural noise tend to deviate more from integer participation.

The same relationship is visible when comparing groups. Fig. 9 shows the distribution of  $z_\xi$  for the “almost-integer” group (top 20% lowest  $|\delta|$ ) and the rest of the dataset. The almost-integer materials have markedly lower noise, with Cliff’s  $\Delta \simeq -0.47$ , a large negative effect size.

#### D. Base-frequency hierarchy across families

The calibrated base frequencies  $f_{\text{base}}^{(k)}$  form a simple, interpretable pattern. Fig. 10 shows the median  $f_{\text{base}}^{(k)}$  per family, ordered by increasing value, together with the global median  $f_{\text{base,med}} \simeq 4.157$ .

Iron-based superconductors cluster near  $f_{\text{base}} \simeq 1$ , suggesting that, in these systems, the mother frequency often participates in the fundamental mode and that  $F_m^*$  itself already behaves as an integer participation number. Heavy-fermion and oxide materials sit near  $f_{\text{base}} \simeq 2$ , consistent with a first harmonic. Binary, molecular and high-pressure superconductors, as well as type-I materials, concentrate around  $f_{\text{base}} \simeq 4$ , the same integer that controls the superfluid–roton ratio in  $^4\text{He}$ . The independently calibrated  $f_{\text{base}}$  for the Superfluid family is  $\simeq 4.0$ , so the global median effectively recovers the  $^4\text{He}$  scale when pooling all families.

### V. DISCUSSION AND OUTLOOK

The first study in this programme showed that a delayed-oscillator locking grammar on the primes

$\{2, 3, 5, 7\}$ , combined with a universal correction law calibrated on classical metals, is sufficient to compress a heterogeneous set of superconductors and superfluid helium into stable prime-space fingerprints. The present work adds a second layer: an integer-participation structure in the renormalised Mother Frequency, once structural noise is taken into account.

On the phenomenological side, the main result is that for each macroscopic family there exists a simple base frequency  $f_{\text{base},k}$  such that the renormalised ratios  $F_m^*/f_{\text{base},k}$  cluster tightly around integers. This clustering is robust under cross-validation and against multiple null models, and it correlates systematically with a structural noise field  $\hat{\xi}$  derived from independent considerations of cluster topology and vector dissipation. Materials that sit closest to an integer participation number tend to exhibit lower structural noise, while strongly non-integer cases concentrate in families where geometric frustration or complex topology are expected.

From a DOFT-inspired viewpoint, these findings are consistent with a picture in which superconducting clusters behave as discrete assemblies of delayed oscillators that seek integer harmonics of a family-specific base frequency, with structural noise quantifying the misfit between the actual geometry and an idealised closed configuration. From a more conventional perspective, the integer structure may simply reflect a coarse-grained count of effective degrees of freedom (layers, chains, or coupled sub-networks) that contribute coherently to the superconducting state.

We deliberately remain agnostic about the microscopic origin of  $f_{\text{base},k}$ . One speculative route is to connect it to known characteristic frequencies in each family, such as dominant phonon branches, spin fluctuations, or collective modes observed in Raman and neutron scattering. Another is to interpret the integer participation as a proxy for cluster geometry, in analogy with magic numbers in metallic clusters, where closed icosahedral shells lead to enhanced stability. At present, the data are not sufficient to distinguish between these scenarios.

The comparison with superfluid  $^4\text{He}$  should also be treated with care. The appearance of a low integer scale linked to the roton minimum is intriguing, but bosonic superfluidity and fermionic superconductivity rely on different microscopic mechanisms. For now we view the helium case as an interesting boundary point for the locking grammar rather than as evidence for a unified oscillator mechanism.

Several extensions are natural. On the empirical side, a larger and more balanced dataset per family would allow sharper estimates of  $f_{\text{base},k}$ , more stringent cross-validation, and a clearer separation between bulk-like and strongly frustrated regimes. On the theoretical side, it would be valuable to embed the integer participation directly into a microscopic delayed-oscillator field theory, or to derive effective  $f_{\text{base},k}$  from simplified models of coupled phonons and electrons. Ultimately, the usefulness of the present construction will be decided by its predictive

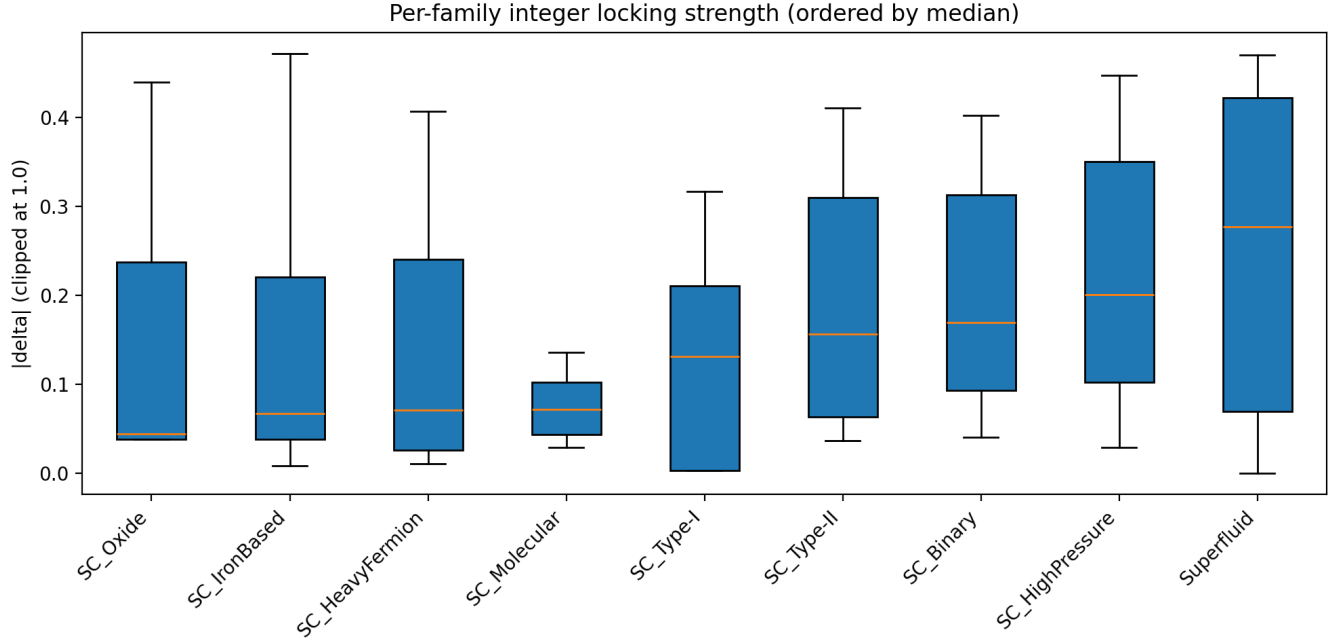


FIG. 7. Per-family distribution of locking residuals  $|\delta|$  (clipped at 1.0). Boxes show interquartile ranges and medians; whiskers extend to 1.5 times the interquartile range. Families are ordered by increasing median residual.

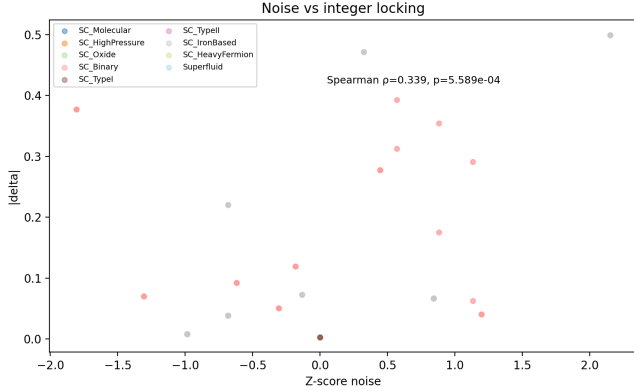


FIG. 8. Absolute locking residual  $|\delta|$  vs structural-noise  $z_\xi$ . Colors indicate material families. The Spearman correlation is  $\rho \simeq 0.34$  with  $p \simeq 5.6 \times 10^{-4}$ , indicating that higher structural noise is associated with weaker integer locking.

power: whether integer participation, combined with the prime-space fingerprints, can help identify promising new superconductors or clarify why certain families systematically fall short of their apparent locking potential.

## VI. CONCLUSION

We have shown that a simple frequency ratio  $F_m = \Theta_D/T_c$ , when corrected for structural noise and rescaled

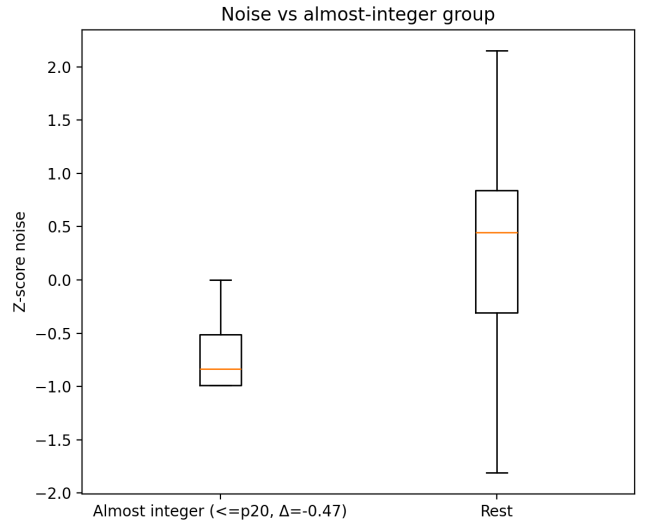


FIG. 9. Structural-noise  $z_\xi$  for materials in the “almost-integer” group (top 20% smallest  $|\delta|$ ) and for the rest of the dataset. The almost-integer group is shifted toward lower noise, with Cliff’s  $\Delta \simeq -0.47$ .

by a family-dependent base frequency, exhibits robust integer participation across a heterogeneous set of superconductors and liquid  $^4\text{He}$ . The effect is statistically strong, structurally modulated and organized by approximate harmonics of a base frequency whose global scale matches that of  $^4\text{He}$ .

Among all families, iron-based superconductors stand

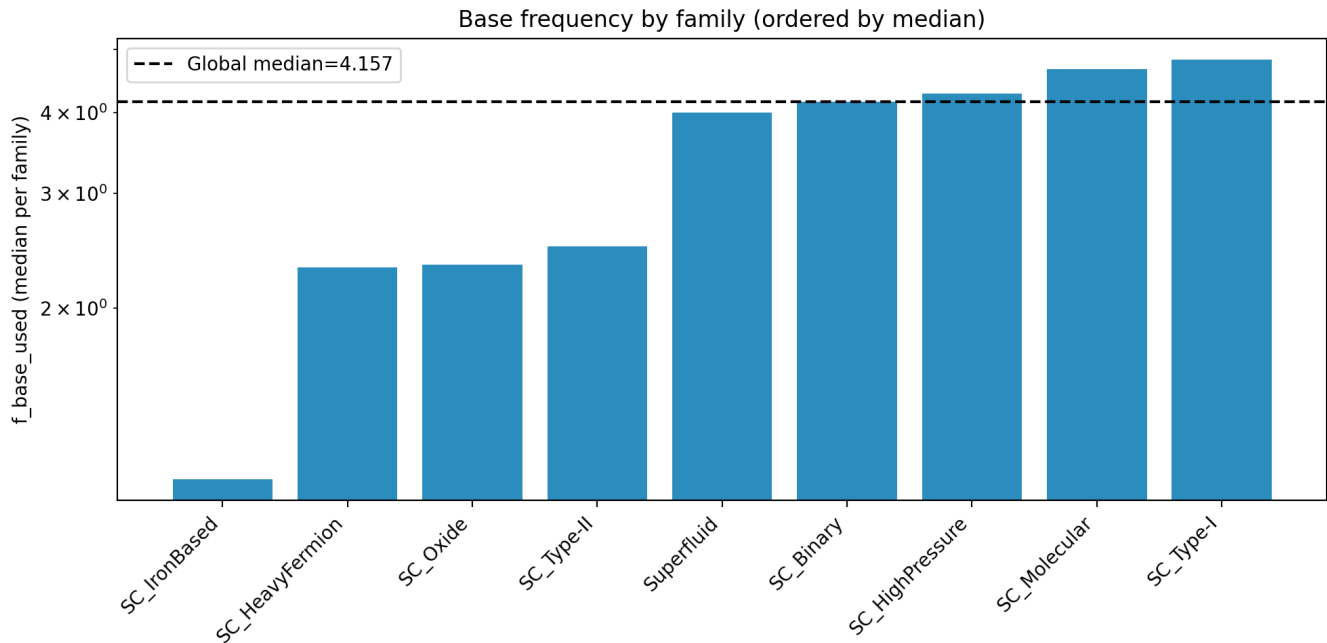


FIG. 10. Base frequency  $f_{\text{base}}$  per family (median over materials), ordered by increasing value. The dashed line marks the global median  $f_{\text{base\_med}} \simeq 4.157$ . Families cluster around approximately harmonic values  $f_{\text{base}} \sim 1, 2$  and  $4$ , with iron-based superconductors near the fundamental and binary / molecular / high-pressure materials near the  $n = 4$  harmonic associated with  $^4\text{He}$ .

out as the clearest realization of this picture: with  $f_{\text{base}} \simeq 1$ , they operate in a natural resonance unit where the noise-corrected mother frequency  $F_{\text{m}}^*$  itself already acts as an integer participation number. In this sense, they behave as ideal, unscaled harmonic oscillators within the phenomenological framework developed here.

Rather than offering a full microscopic theory, our goal is to provide a compact phenomenological summary of patterns that any future theory ought to explain. The framework—mother frequency, structural noise, base-frequency calibration and integer participation—is deliberately minimal and can be applied to expanded datasets as they become available.

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AI tools were used as assistants in the development of the code and this manuscript. In particular, OpenAI GPT-5.1 Thinking, Google Gemini Pro, and Anthropic Claude were employed for drafting text, refactoring analysis pipelines, and checking statistical procedures. All modelling decisions, dataset curation and interpretations

presented here were made and verified by the human author.



# APPENDIX: STRUCTURAL-NOISE DIAGNOSTICS

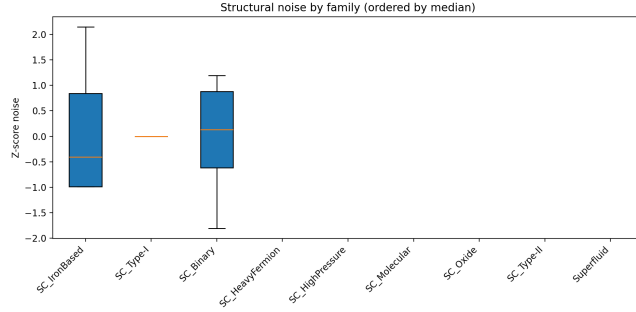


FIG. 11. Structural-noise  $z$ -score  $z_\xi$  by family (only families with a non-degenerate noise model are shown). Distributions are standardized within each family; boxes show interquartile ranges and medians.

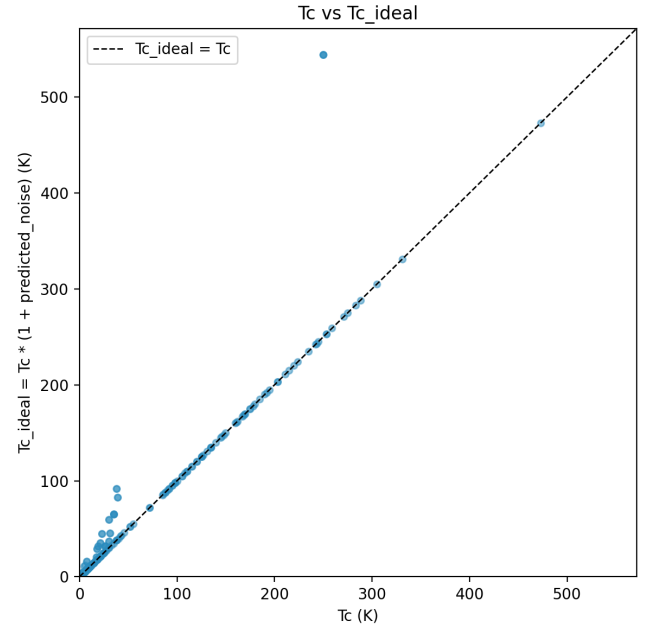


FIG. 12. Measured transition temperature  $T_c$  vs ideal temperature  $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$ . The dashed line indicates  $T_c^{\text{ideal}} = T_c$ . Most materials lie close to the identity, with a subset receiving larger upward corrections due to structural noise.

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